

Linux Administration Interview Questions (21–40)

21. What is the difference between TCP and UDP?

TCP is connection-oriented, reliable, slower, used for web/SSH. UDP is connectionless, unreliable, faster, used for streaming/DNS.

22. How do you troubleshoot network connectivity in Linux?

Check IP (ip addr), ping localhost/gateway, traceroute, check DNS (dig/nslookup), check ports (ss/netstat), capture packets (tcpdump).

23. Explain the difference between SAN, NAS, and DAS.

SAN: block-level, high performance; NAS: file-level, shared via NFS/SMB; DAS: directly attached storage like HDD/SSD.

24. What is LVM? How do you create and extend a logical volume?

LVM = Logical Volume Manager for flexible storage. Create: pvcreate, vgcreate, lvcreate, mkfs, mount. Extend: lvextend + resize2fs.

25. Difference between ext3, ext4, and xfs.

ext3: legacy, 2TB files. ext4: default, 16TB, better performance. xfs: high performance, good for DB/large files.

26. What is RAID? Explain RAID levels.

RAID = Redundant Array of Independent Disks. RAID 0: striping (fast). RAID 1: mirroring. RAID 5: parity. RAID 10: stripe+mirror.

27. What is SELinux/AppArmor?

SELinux = security policies with MAC. AppArmor = profile-based restrictions. Both restrict processes beyond file permissions.

28. What is iSCSI, and where is it used?

iSCSI = block storage protocol over TCP/IP, used in SAN/datacenters.

29. Explain the difference between init and systemd.

init: legacy, sequential, slow. systemd: modern, parallel startup, journald, unit files, targets.

30. How do you mount NFS and CIFS shares?

NFS: mount -t nfs server:/share /mnt. CIFS: mount -t cifs //server/share /mnt -o username=user.

31. A user cannot log in — how do you troubleshoot?

Check account (id user), /etc/passwd, /etc/shadow, password expiration (chage), and logs (/var/log/auth.log).

32. A server is running slow — what steps will you take?

Check CPU/mem (top, htop, free), disk I/O (iostat, iotop), load (uptime), network (iftop), logs.

33. Disk space is full — how will you find and clear space?

Check df -h, find large files (du -sh * | sort), clean logs, remove cache, use logrotate.

34. A service is not starting — how will you debug?

Check status (systemctl), logs (journalctl), config errors, manual start.

35. The network is unreachable — what troubleshooting steps will you follow?

Check interface (ip link), IP (ip addr), ping gateway/host, routes (ip route), firewall (iptables).

36. The server is not booting — how do you fix it?

Use GRUB recovery, rescue mode, fsck, rebuild initramfs, grub-install.

37. Logs are filling up the disk — how will you handle it?

Check log size (du), configure logrotate, compress/archive logs, redirect debug logs.

38. You suspect high I/O wait — how will you confirm and fix it?

Check with iostat/vmstat/sar. Fix by optimizing apps, faster disks (SSD), RAID tuning.

39. How do you recover a deleted file in Linux?

If open: lsof | grep deleted. Use extundelete for ext FS. Restore from backups.

40. How do you secure SSH on a Linux server?

Change port, disable root login, use key-based auth, firewall rules, Fail2Ban.